Vel Tech
Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
University
Avadi, Chennai

SCHOOL OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT

Name	: Jarrpana Sanjeev Varma
VTU Number	: VTU27899
Course Code/ Course Name	: **10212CS211/Artificial Intelligence Techniques**
Year	: Summer 2026-27
Course Handling Faculty	: **Dr.S.Pragadeeswaran TTS3903**
Slot	: **S6L1**
Course Coordinator	: **Dr. R. RoselinKiruba TTS3227**
Assignment Group Number	: 01
Date	: 26/10/2026

Assignment Mark Allocation

Criteria	Total Mark	Mark Achieved
Concept Understanding (A)	1	
Accuracy & Correctness (B)	1	
Presentation Clarity & Communication (C)	1	
Relevance & Completion (D)	1	
Timeliness (E)	1	
Total (A+B+C+D+E)	5	

General Comments:

Faculty Signature

Faculty Name & TTS:

CO NOs	Course Outcomes	Level of Learning domain(Based on revised Blooms taxonomy)
CO1	Identify the problem-solving techniques by using State Space search and heuristic search approach	K3
CO2	Use evolutionary and advanced search techniques to provide optimized solution.	K3
CO3	Solve unified planning approach and Constraint-satisfaction problems.	K3
CO4	Apply Logical knowledge representation for rules and fact-based approach.	K3
CO5	Build the logic of Uncertainty and Reasoning using resolution methods.	K3

Question Group:

GROUP -1			
S.No	List of Questions	CO	K Level
1	Apply BFS and DFS to solve given tree	CO1	K2
2	Apply A* Algorithm to find sortest path.	CO2	K3
3	Choose an Interesting problem for real-world.	CO3	K2
4	Express following sentence a) A person is b/w rock & hard place Tom Believe many want to marry a sailor	CO4	K2
5	Consider Axioms ① Everyone love Santa ② Everyone who love santa love any reindeer ③ Rudolph's reindeer & Rudolph has red nose	CO5	K3

1. Apply BFS And DFS :-

From the Diagram take node 1 as the starting node and visit adjacent nodes from left to right.

Breadth First Search:-

BFS visits nodes level by level using a queue Traversal sequence:-

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8$

Depth First Search:-

DFS explore are branch as far as possible before backtracking

Traversal sequence:-

$1 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 8 \rightarrow 7 \rightarrow 4 \rightarrow 5$

2. Apply A^* Algorithm:-

A^* uses evaluation function

$$f(n) = g(n) + h(n)$$

Where

$g(n)$ = actual cost

$h(n)$ = heuristic goal node

$f(n)$ = estimated cost

from · Graph

S - B = 5

B - E = 6

E - F = 4

F - G = 3

steps :-

1. start at node S

2. Expand the promising node using
   $f(n) = g(n) + h(n)$

3. select path
   $S \rightarrow B \rightarrow E \rightarrow F \rightarrow G$

4. Calculate cost

   $5 + 6 + 4 + 3 = 18$

Shortest Path

$S \rightarrow B \rightarrow E \rightarrow F \rightarrow G$

Total · path = 18

3. College Student Travel Planning

Initial state:-

A student reach is at home and has to reach college.

Goal state:-

The student reaches college on time

Actions:-

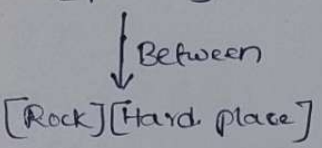
1. Check the bus or train
2. Choose the best route
3. Buy a ticket.
4. Travel to college

Planning Method:-

Goal state planning (or) A* search can be used to find suitable route.

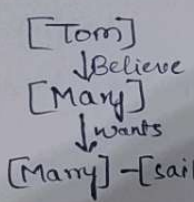
4.

a) A person is Between rock and hard place Conceptual Graph :-



Meaning :- A person is in a difficult situation with limited choices.

b) Tom believe that mary wants to marry a sailor



5.

Step 1 :- Convert into predict logical classes

1. ¬ child (x) ∨ loves (x, santa)
2. ¬ loves (x, santa) ∨ ¬ Remainder (y) ∨ loves (x, y)
3. Reinder (Rudolph)
4. Red Nose (Rudolph)
5. ¬ Red Nose (x) ∨ weird (x) ∨ clown (x)

6. Reindeer (x) ∨ ¬clown (x)

7. ¬weird (y) ∨ ¬loves (scrooge, y)

8. child (scrooge)

Step 2:-

From clauses (1) & (8)

love (scrooge, santa)

from clauses (2), (3).

above result

loves (scrooge, Rudolph)

from clauses (4)(5)

weird (Rudolph) ∨ clown (Rudolph)

From clause (3) & (6)

¬ clown (Rudolph)

Resolving these clauses gives:

weird (Roudolph)

from clauses (7):

¬ loves (scrooge, Rudolph)

But we already desired:

loves (scrooge, Rudolph)

Therefore we obtain in a contradiction